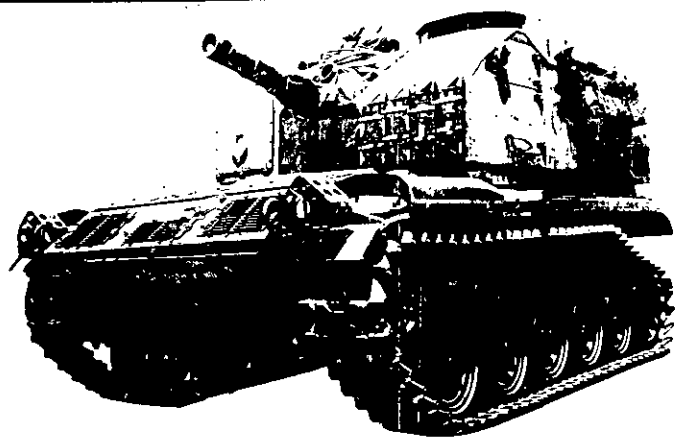
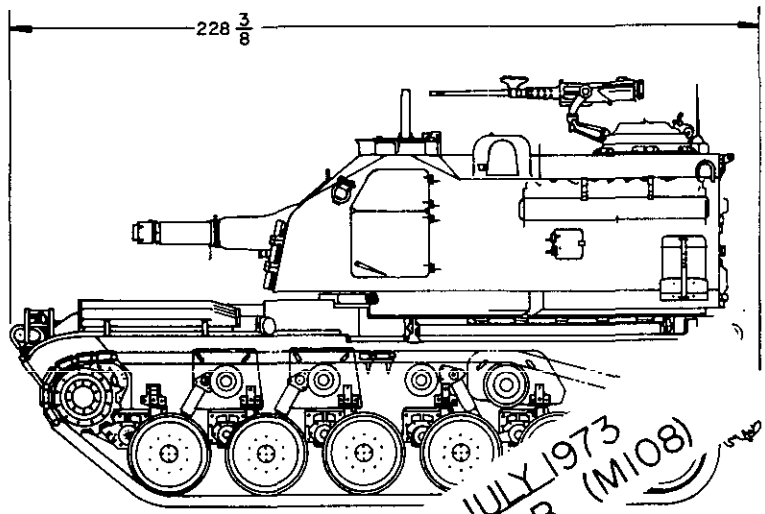
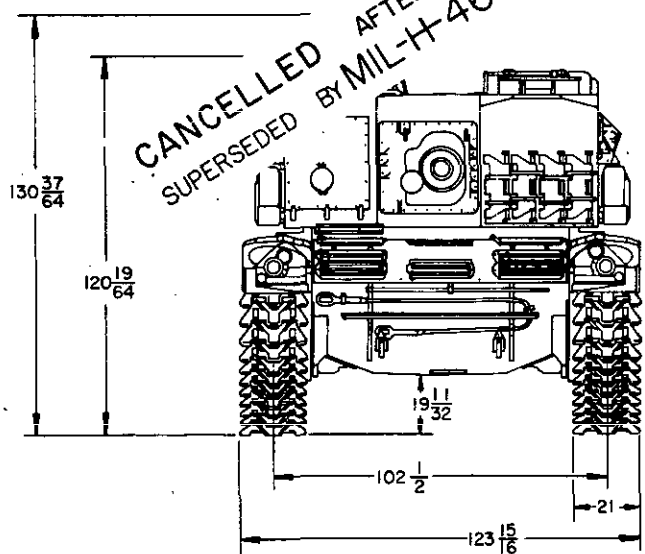
REVIEW ACTIVITIES: ARMY—WC, MU, EL

FED. SUP CLASS
2350





CANCELLED AFTER 25 JULY 1973
SUPERSEDED BY MIL-H-46709B (MI08)



DIMENSIONS ARE IN INCHES AND ARE SHOWN FOR ENGINEERING REFERENCE ONLY

MS PART NO.	DRAWING NO.
MS 51985—1	8736128

APPROVED 7 MAY 1964 REVISED (A) 25 JULY 1973

This standard has been approved by the Department of the **ARMY** and is mandatory for use by that activity. All other military activities are required to apply this standard where suitable.

P.A. MO (ATAC) Other Cust	TITLE HOWITZER, LIGHT, SELF PROPELLED 105 MM, M52A1	MILITARY STANDARD MS 51985 (MO)
PROCUREMENT SPECIFICATION NONE	SUPERSEDES:	SHEET 1 OF 2

DD FORM 1 SEP 67 672-1 (Limited coordination)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

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VEHICLE CHARACTERISTICS

FED. SUP CLASS 2350

REVIEW ACTIVITIES: ARMY – WC, MU, EL

GENERAL

WEIGHT (COMBAT LOADED) APPROX. 53,000 LBS.
CREW 5
WEIGHT (LESS CREW, STOWAGE AND FUEL) 49,800 LBS.
CENTER OF GRAVITY 44.5 IN. ABOVE GROUND
98.2 IN. FROM CENTERLINE OF SPROCKET
UNIT GROUND PRESSURE 8.5 LBS/SQ. IN.
TURRET RING DIA. (INSIDE) 73 IN.

ARMOR

VISION AND FIRE CONTROL EQUIPMENT

PERISCOPE, VISION	TYPE M13	QTY. 4
PERISCOPE, COMMANDERS	TYPE M15A1	QTY. 1
PERISCOPE, DRIVERS	TYPE M17	QTY. 5
PERISCOPE	TYPE M23	QTY. 1
SIGHT, TELESCOPE PANORAMIC	TYPE M100	QTY. 1
SIGHT, TELESCOPE	TYPE M101	QTY. 1
QUADRANT, GUNNERS	TYPE M1A1	QTY. 1
SETTER, FUZE	TYPE M26	QTY. 1
SETTER, FUZE	TYPE M27	QTY. 1

ARMAMENT, PRIMARY

HOWITZER, 105 MM, M49 MOUNT, M85
RECOIL MECHANISM TYPE: CONCENTRIC HYDRO-SPRING
TRAVERSE: RIGHT, 60° LEFT, 60°
POWER TYPE: NONE MANUAL TYPE: GEAR & PINION
ELEVATION: +65° DEPRESSION: -10°
POWER TYPE: NONE MANUAL TYPE: RACK & PINION

ARMAMENT, SECONDARY

GUN, MACHINE, CAL. .50 M2, A.A. MOUNT, CAB ROOF
TRAVERSE, 360° ELEVATION +85°
DEPRESSION -15°
GUN, SUB-MACHINE, CAL. .45 M3A1 (1) IN CAB
CARBINE, CAL. .30, M2 (4) IN CAB

AMMUNITION

102 RDS. 105 MM
945 RDS. CAL. .50
840 RDS. 30 CAL.
180 RDS. 45 CAL.
8 GRENADES, HAND

RUNNING GEAR

SUSPENSION TYPE: INDEPENDENT TORSION BAR WITH TRAILING IDLER
NO. OF WHEELS: 5 DUALS EACH SIDE, DISK TYPE WHEEL SIZE 25 1/2 IN.
TIRES, 10 DUALS; TYPE, SOLID RUBBER, 25 1/2 x 4 1/2
TRACK: TYPE T91E3, STEEL, W/DETACH, RUBBER PADS
NO. SHOES/TRACK, 74L, 75R PITCH, 6 IN.
WIDTH, 21 IN.

ELECTRICAL SYSTEM

NOMINAL VOLTAGE, 24
GENERATOR, MAIN, 150 AMP, PART NO. 8724633
GENERATOR, AUX. NONE
BATTERY: (4) TYPE 6TN, 12-V, 100 AMP., MS35000-4
IGNITION SYSTEM: MAGNETO, MS 51333
COIL, MS 51355-1

POWER PACKAGE

ENGINE

TYPE: 6 CYL. HORIZONTAL OPPOSED, AIR COOLED, FUEL INJECTION, MS 51314-1
DISPLACEMENT, 895 CU. IN. BORE, 5.75 IN.
STROKE, 5.75 IN.
GOVERNED SPEED, 2800 RPM COMPRESSION RATIO 5.5:1
FUEL, GASOLINE RATING, 80 OCTANE
BRAKE HORSEPOWER (STANDARD CONDITIONS) 500 @ 2800 RPM
TORQUE (STANDARD CONDITIONS) 945 LB./FT. @ 2000 RPM
MAIN COOLING SYSTEM: AIR, ENGINE DRIVEN INTEGRAL FAN
OIL COOLING SYSTEM: AIR, ENGINE DRIVEN INTEGRAL FAN

POWER TRAIN

TYPE: CROSS DRIVE TRANSMISSION, PART NO. 7527947
HYDRAULIC CONVERTER, SINGLE STAGE, MULTIPHASE STALL MULTIPLICATION, 4:1
OVERALL USABLE RATIOS: LOW, 14.9
HIGH, 3.9; REVERSE, 14.9
STEERING CONTROL: HYDRAULIC SERVO, HANDLE BAR STEERING RATE, 6.8 RPM
TURNING RADIUS, PIVOTS
BRAKES: MULTIPLE DISK, OPERATION: FOOT PEDAL, HYDRO-MECHANICAL
OIL COOLING SYSTEM: AIR, ENGINE DRIVEN INTEGRAL FAN

FINAL REDUCTION

TYPE: SPUR GEAR, RATIO 4.75:1 FT/REV 6
SPROCKET PITCH DIA. 23.422 NO. OF TEETH 12

COMMUNICATIONS

RADIO SET: SCR-619 OR SCR-610
LOCATION: CAB
RADIO SET: AN/PRC 8, 9, OR 10
INTERPHONE: MODEL AN/UIC-1, 5 OUTLETS
EXTERNAL INTERPHONE EXT. KIT – MODEL C-980/U

FIRE EXTINGUISHER

FIXED: (2) CO_2, 10 LB. BOTTLES
PORTABLE: (1) CO_2, 5 LB. BOTTLE

CAPACITIES

FUEL, TOTAL, 174 GALLONS
CRANKCASE, REFILL, 44 QUARTS
TRANSMISSION, REFILL, 44 QUARTS
FINAL DRIVE, REFILL, 5 QUARTS

PERFORMANCE

GROSS HORSEPOWER TO WEIGHT RATIO: 18.9 HP/TON
MAXIMUM TRACTIVE EFFORT, 41,000 LBS.
TE/W, 0.77
MAX. SPEED 42 MPH MAX. GRADE 60%
MAX. TRENCH 72 IN. MAX. VERTICAL WALL 30 IN
MAX. FORDING DEPTH 48 IN.
CRUISING RANGE 100 MILES

THIS VEHICLE SHALL BE IN ACCORDANCE WITH REFERENCED DRAWING NO. 8736128, COPY OF WHICH MAY BE OBTAINED FROM ARMY TANK AUTOMOTIVE CENTER, DETROIT ARSENAL, WARREN, MICHIGAN, ATTN: STANDARDIZATION BRANCH.

APPROVED 7 MAY 1964 REVISED (A) FOR CHANGE SEE SHEET 1

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